

Cubic Gymnastics

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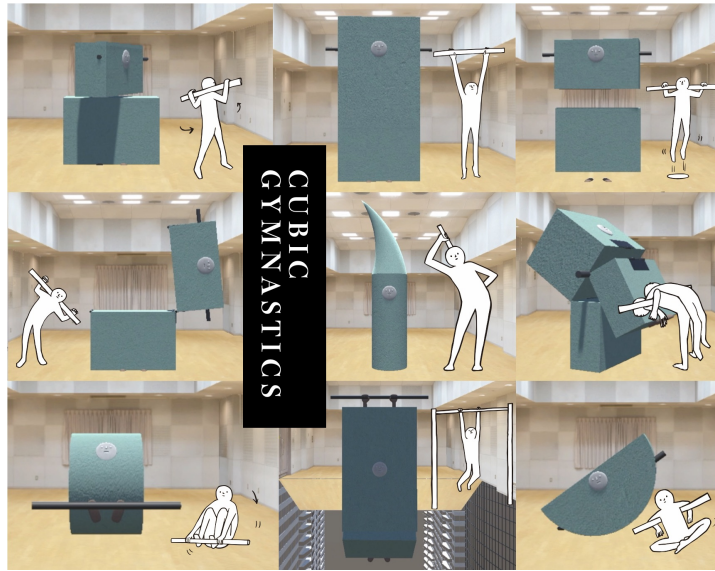


Figure 1: CUBIC GYMNASTICS

ABSTRACT

"Cubic Gymnastics" is an XR system that enables users to engage in a wide range of exercises specifically designed for geometric shapes, which are visualized within the HMD space. By wearing the HMD, users can mentally immerse themselves in a geometrical body viewed in the mirror referred as the "cubic body," comprised of multiple scalable cuboids and/or curved solid shapes. The physical restriction imposed on the user's body by a long rod, typically limiting hand motion, creates a perception of being transformed into a flat form, thereby enhancing the illusion of owning the cubic body. These geometric exercises involve unique movements that are physically impossible for the human body, including the separation of the upper and lower body, adhesion, stretching, deformation, and rotation-on-the-ground. Results from a survey conducted during a laboratory exhibition confirmed that a significant number of

participants strongly experienced the sensation of performing these unconventional object-like movements specific to the transformed geometric shapes.

ACM Reference Format:

Kenri Kodaka, Kousuke Motohashi, and Tsuyoshi Suzuki. 2023. Cubic Gymnastics. In *SIGGRAPH Asia 2023 XR (SA XR '23)*, December 12-15, 2023. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610549.3614605>

1 BACKGROUND

It is widely recognized as "full-body illusion" that can make individuals feel as if they possess a different body than their actual physical form[Pyasik et al. 2022]. In the case of first-person perspective full-body illusion, a mirror is placed within the HMD space to synchronize the user's sense of movement with the visual motion of the avatar or the mannequin body[Preston et al. 2015]. This allows the individual to experience a sense of their own body being replaced by the body of the avatar. This approach is widely used in the world of social VR, including platforms like VRChat.

In the context of full-body illusions, such as those found in typical social VR experiences, the avatars used often maintain the anatomical structure of a human body. This is done to create a sense of familiarity and facilitate social interaction. As a result,

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SA XR '23, December 12-15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0316-4/23/12.

<https://doi.org/10.1145/3610549.3614605>

these avatars are not typically designed to simulate object-like movements that involve independent movement of the upper and lower body elements, rotation on the ground, and similar actions.

However, it is important to note that unconventional movements, such as those involving object-like motions, have the potential to greatly enhance the freedom of actions in the metaverse space. For example, the separation of the upper and lower body in VR space has the potential to significantly alleviate the constraints of movements that would typically be expected in physical space. Thus, by introducing movements that go beyond the limitations of the human body, users can experience a heightened sense of exploration, creativity, and interaction within virtual environments. These unconventional movements can add a new level of immersion and expand the possibilities of what can be achieved in the metaverse.

Based on these insights, we introduce the "Cubic Gymnastics" HMD system. This system is specifically designed to provide users with a unique experience of owning a geometrical body and engaging in unconventional, object-like movements.

2 DESIGN

The essential design of the interactive system for "Cubic Gymnastics" is very simple. The user wears an HMD (such as the HTC Vive Pro) and holds a long rod in various forms, with a vive tracker attached to the edge of the rod. The long rod typically serves as a means to restrict the movement of the user's body during specific gymnastics exercises, creating a planar object-like form.

The Cubic Body is viewed through a mirror placed in front of the user in the HMD space. The elements of the Cubic Body include various shapes such as cubes, cones, cylinders, and half-cylinders. These elements can be combined in different configurations to form the Cubic Body, with each element being separable from one another. The variables (position, orientation, scale) of each geometric element are determined by the positional information of the head and the fixing rod, either individually or in combination.

"Cubic Gymnastics" can be broadly classified into five geometric movements: separation and adhesion, hinge motion, flexural motion, curved-surface grounding, and pendulum.

2.1 separation and adhesion

In the physical space, the user is equipped with a long rod positioned on their back as the starting point. They have the freedom to perform a range of movements, such as twisting their upper body from side to side, squatting, jumping, and even raising the rod in a cheering motion. In the HMD space, the upper and lower cuboids can move independently. For instance, when twisting the upper body, both cuboids share a common axis of rotation, but only the upper cuboid rotates in the desired direction. Similarly, squatting or cheering causes the upper and lower cuboids to connect and stretch in the intended direction. Additionally, during a jump, the upper cuboid pops and separates from the lower cuboid.

2.2 hinge motion

The user, while keeping the long rod fixed on their back, stands upright and tilt their upper body forwards, backwards, or sideways. Alternatively, they can lie down and perform abdominal exercises

by lifting only their upper body, similar to a crunch motion. In the HMD space, in response to such movements, only the upper block tilts forward, backward, left, or right. During this movement, the adjacent region between the upper and lower blocks is visually represented as a black hinge.

2.3 flexural motion

The user holds the long rod vertically against their back. In this position, when the body is tilted to the left or right, in the HMD space, the upper cone bends significantly in the corresponding direction. Similarly, when squatting while maintaining this position, the cylindrical shape is compressed to a lower height.

2.4 curved-surface grounding

When the user lowers their hips to the ground, bends their knees, and keeps the long rod hooked onto their feet, then rotates backward, in the HMD space, the cylinder smoothly rotates in the corresponding direction. Additionally, while keeping the long rod fixed to the back, in a cross-legged position, when the body sways from side to side, in the HMD space, a solid shape resembling a half-cut cylinder oscillates left and right, with the curved surface as the ground.

2.5 pendulum

The user hangs on a horizontal bar and sways their body back and forth. In the HMD space, the lower block of the two blocks bends at the knee area, while the upper block swings back and forth like a pendulum. For this interaction, the Vive Tracker is attached to the user's waist position.

3 REACTIONS TO THE EXHIBITION

This system was opened to the public at a laboratory exhibition in November 2022. Typically, participants are given the opportunity to experience various Cubic Gymnastics exercises in a sequential manner for approximately 10 minutes. After the experience, participants were asked to rate the sensation of detachment between the upper and lower body during several hinge motions and the sensation of separation between the upper and lower body during jumping on a scale of -3 to +3 in a questionnaire.

The results show that out of 66 participants, 47/45 agreed with the illusory sense of detachment during the sideways/back-and-forth hinge motions, rating it positively with a score of +1 or higher. Furthermore, 49 out of 66 participants positively agreed with the illusory sense of separation between the upper and lower body. Overall, the technology consistently created the perception of owning a geometrical body and successfully elicited the sensation of performing object-like motions.

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